



ABS IN INDIA

Rules | Specific requirements | Practical considerations | Contacts

International agreements on ABS

- Convention on Biological Diversity (CBD): Party since 19 May 1994
- Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits resulting from their Utilization (ABS): Party since 12 October 2014
- International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA): Party since 29 June 2004

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Introduction

India is a megadiverse country, which harbors nearly 8% of all recorded species – as well as 18% of the global population – on less than 2.5% of world's land. Integrating ecological, economic and social goals has been described as a “delicate balance.” India is also recognized as one of the planet's main centers of origin for the agricultural crops, and local farmers still play an important role in conserving traditional varieties, as well as in breeding new ones. Moreover, India has a vast heritage of traditional knowledge associated to the properties and applications of biological resources. For example, 6560 species of Indian medicinal plants have been recorded across codified systems such as Ayurveda and the folk traditions of local communities.

India has been an active participant in international efforts for the conservation and sustainable use of biodiversity, as well as the fair and equitable sharing of resulting benefits. It is a party to the Convention on Biological Diversity (CBD) and the Nagoya Protocol. It has also pioneered efforts such as the Traditional Knowledge Digital Library, a digital database developed to prevent misappropriation of traditional knowledge by providing access to related information to patent offices around the world.

The competent national authority on access and benefit sharing (ABS) has issued over 900 authorizations for research, commercial utilization and other activities linked to biological resources. Nevertheless, the level of awareness and compliance among companies in India and abroad remains limited. There are uncertainties on how basic concepts are interpreted and how requirements work in practice. This note summarizes key points in Indian ABS rules and their practical implications – particularly for companies innovating on and sourcing natural ingredients for food, cosmetics and natural pharmaceuticals.

ABS in national laws and regulations

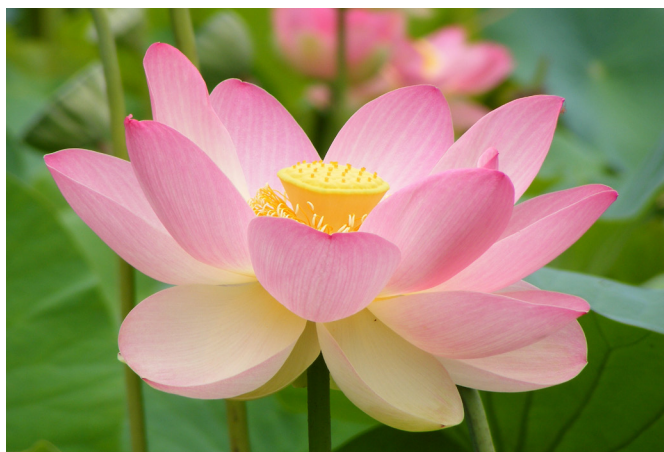
- The Biological Diversity Act (2002), with ABS requirements in force since 1 July 2004
- The Biological Diversity Rules (2004), in force since 1 July 2004
- Guidelines on Access to Biological Resources and Associated Knowledge and Benefits Sharing Regulations (2014), in force since 21 November 2014
- Notification on Exemption of Crops listed in the Annex I of the International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA), in force since 17 December 2014
- Guidelines for Implementing the International Treaty of Plant Genetic Resources for Food and Agriculture for Facilitated Access under the Multilateral System, adopted by Department of Agriculture and Cooperation on 30 July 2014
- Notification on biological resources normally traded as commodities (2016), in force since 7 April 2016, with amendments in force since 7 November 2017
- Guidelines on Upfront Payments (not-dated)

Scope

Indian rules on biodiversity regulate access to any biological resource occurring in India or associated traditional knowledge, for the purposes of research, characterization or commercial utilization. That is, ABS requirements in India have a broad scope, covering a wide range of both resources and activities.

Biological resources

Biological resources are defined as “plants, animals, and microorganisms or parts thereof,” as well as “their genetic material and by-products.” The term “by-products” would cover biochemical compounds – even if such compounds have undergone physical or chemical processing. Indian requirements cover not only species native to India, but also exotic plants or other biological resources that have been obtained from the country.



Human genetic resources are not included. Additionally, there are two important exceptions to biological resources subject to ABS requirements:

- **Plants normally traded as commodities** (commonly referred to as NTACs). The Ministry of Environment, Forest and Climate Change (MoEFCC) has notified a list of over 400 plant species, from crops to medicinal plants, that are widely traded and, therefore, excluded from ABS requirements. The NTACs exception only applies if the company is commercializing the specific plant part and type of plant (e.g. cultivated or wild) in the list. To be excluded, commercialization of the specified plant part must also be for “common practice” – for the purposes the plant is generally traded and processed. For example, processing avocado fruit for use in food products is likely to be considered common practice, but using the avocado fruit to obtain a cosmetic extract would likely not be, and thus not be exempted from ABS. Moreover, the NTACs exception does not apply if plants are used for research or in relation to an application for intellectual property protection.
- **Value-added products** (commonly referred to as VAPs) are defined as products that have portions or extracts of biological resources “in unrecognizable and physically inseparable form.” That is, for this second exception, there is no list but rather a set of criteria. The term “products” is seen to limit this exception to consumer products – that is, products that have undergone all stages of production and are packaged in their final container. It would not cover intermediate products, inputs or ingredients.

Specific questions on scope

- **Native and non-native plants.** No distinction is made in ABS-related rules between native or non-native plants. As long as biological resources are obtained in India, they are considered to fall within the scope of ABS requirements.
- **Microbes.** The definition of biological resources includes a specific reference to microorganisms, thus bringing them into the scope of ABS requirements.
- **DNA samples.** The definition of biological resources includes a specific reference to genetic material, thus bringing DNA samples into the scope of ABS requirements.
- **Ex-situ access.** The Biological Diversity Act and Rules have no specific provisions on ex-situ access. However, ex-situ access is implied through the use of terms such as “obtain” and “occurring in India” in Section 3(1). ABS requirements thus apply to cases of access of biological resources from collections, seed banks, gene banks, botanical gardens, etc.

Research, commercial utilization and other relevant activities

Prior approval is required to use biological resource or associated traditional knowledge for “research,” “bio-survey and bio-utilization,” and “commercial utilization.” The terms are specifically defined, yet involve activities that, in practice, often overlap. What is important is recognizing that most activities along supply chains will be covered – including sourcing biological material to produce inputs or ingredients for drugs, industrial enzymes, fragrances and cosmetics – and require applying for authorization using the same form (see below).

Other activities covered by ABS requirements include:

- Applications for intellectual property rights, within or outside India, for inventions based on any research or information on a biological resource obtained from India
- The transfer of research results, particularly to persons not Indian citizens or companies not incorporated in India or with foreign participation in their shares or management
- The transfer of biological resources accessed under ABS rules to any other persons or organizations

Another important point is that applications may only refer to upcoming activities. Biodiversity rules in India do not enable competent authorities to retroactively approve access to biological resources or associated traditional knowledge. MoEFCC periodically enables the National Biodiversity Authority (NBA) to consider applications to regularize past activities, for a limited time period.



Defining competent authorities on ABS

Indian citizens and companies incorporated in India with no foreign participation in their shares or management apply for ABS permits before the State Biodiversity Boards, which have now been established in all of India’s 29 states. Other persons or companies apply for ABS permits before the National Biodiversity Authority (NBA), which is the competent national authority and is based in Chennai. The NBA involves state and local level authorities in its consultations.

Both state and national-level ABS procedures are based on the Indian Biodiversity Act and Rules, as well as complementary notifications and guidelines. Recent court decisions have clarified that ABS requirements are equally applicable in case of Indian persons and companies. However, in practice, certain interpretations and procedures may differ according to the competent authority involved.

Access requirements

In India, an authorization is required prior to the **utilization** (rather than the access) of biological resources and associated traditional knowledge, as per the activities listed in the Biodiversity Act as triggering ABS requirements. The authorization is granted through mutually agreed terms – a contract that outlines permitted uses, establishes terms and conditions for research or other activities and defines the mechanism and amount for benefit sharing – agreed upon between the applicant and the competent authorities, on behalf of the providers.

The process is launched through an application to the competent authorities (see box). It is a single window application process. Requirements for selected applications are described in Table 1. The applicant is not responsible for negotiating prior informed consent with local actors. It is the competent authorities that, having received an application, are charged with consulting relevant

stakeholders, as a way of formalizing the PIC process. For example, the NBA involves the relevant State Biodiversity Board (SBB), which in turn holds consultations with the Biodiversity Management Committee in the area where biological resources will be collected or harvested.

At the national level, once an application is completed and local consultations concluded, it is forwarded to the Expert Committee on Access and Benefit Sharing, which is comprised of over 25 experts and representatives from government entities and academic institutions. The expert committee reviews the application and advises on the terms and conditions, including on fair and equitable benefit sharing. Based on these recommendations, the NBA proposes and agrees upon mutually agreed terms with the applicant. The approval for access to biological resources is then granted and communicated to the ABS Clearing House. In the past year, approvals have generally taken between six months to one year. On approval, the user is required to submit an annual report to the competent authorities.

Table 1. Selected applications for access to biological resources

Proposed activities	Application	Information required includes:	Fee
Research	Online application using form I	- Description and quantity of biological resources and any associated traditional knowledge - Area where collection or harvest will take place - Person accessing on behalf of applicant - Purpose of access - Expected economic and other benefits for applicant - Proposed arrangements for benefit sharing	10'000 INR
Commercial utilization	Online application using forms I and A	Additionally, form A requires information on supplier, price paid for biological resources and prospective buyers, if known.	10'000 INR
Transfer of research results	Online application using form II	- Details of the research results - Expected economic and other benefits for applicant and organization that would receive the results	5'000 INR
Intellectual property application	Online application using form III	- Description of invention - Biological resources or associated traditional knowledge used - Location of collection or harvest/ information of identified individual or community holding traditional knowledge - Entity that conducted R&D - Expected economic and other benefits for applicant	500 INR
Transfer to third parties	Online application using form IV	- Purpose of the transfer of biological resources - Details of agreement between applicant and third party	10'000 INR

Benefit-sharing requirements

Mechanisms and arrangements for benefit sharing are established in the mutually agreed terms signed by applicants with the competent authorities. However, since 2014, parameters for benefit sharing arrangements in the case of commercial utilization of biological resources and intellectual property applications have been established through regulations entitled “Guidelines on Access to Biological Resources and Associated Knowledge and Benefits Sharing” (see Table 2). It is important to note that discussions are underway to revise these guidelines. Once revisions are adopted, this fact sheet will be updated online.

Additionally, the NBA has issued guidelines defining the amount of upfront payments required for access to biological resources for research activities, based on the type of biological resources, the quantity collected, the sector in which research is taking place and any access to associated traditional knowledge. For example, the wild collection of 5 kg of plants for research into cosmetic applications would be subject to a payment of 20'000 INR. The payment for accessing 500 ml of industrial effluent for research into environmental remediation would amount to 1'000 INR.

Table 2. Benefit sharing options and parameters

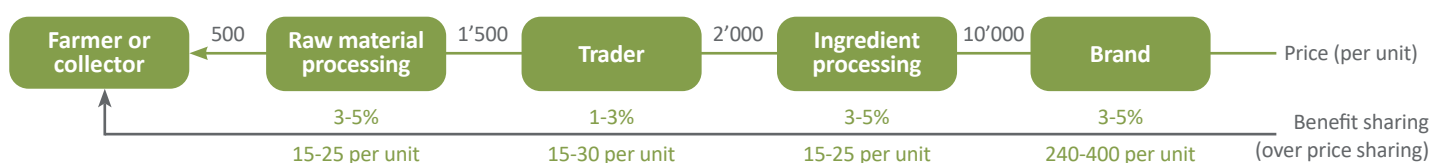
Proposed activities	Options for benefit sharing	Parameters for benefit sharing	Considerations
Commercial utilization	Percentage of purchase price of biological resources	Trader.....1-3% Manufacturer.....3-5%	Specific amount is proposed by NBA, considering activities, potential markets, investments and technologies involved, in line with recommendations from expert committee.
	Percentage of net ex-factory price of resulting product	Depending on total amount, 0.1-0.5%	
Intellectual property	For own exploitation, percentage of net ex-factory price of resulting product	Depending on sector, 0.2-1.0%	
	For licensing, percentage of fees and royalties received	• 3-5% for fees • 2-5% for royalties	

Benefits should be aimed to promote local livelihoods and conservation and sustainable use of biodiversity. Payments are made to the competent authorities, which nevertheless only retain 5% of total benefits shared, to cover administrative fees. The remaining 95% is transferred to the providers of biological resources or associated

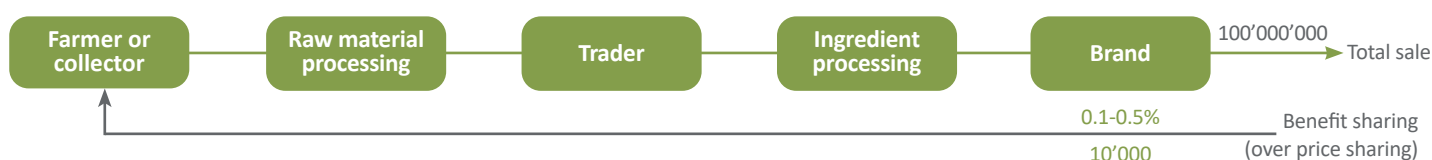
traditional knowledge. If it was not possible to identify these providers, then benefits would be invested in projects in the relevant areas. It is also possible to share non-monetary benefits. However, this is an option that in practice has not yet been explored.

Hypothetical examples of benefit sharing along the supply chain

On price of biological material



On ex-factory sale





Sanctions and compliance

The Biological Diversity Act and Rules include provisions on enforcement and penalties. At the state level, notices and fines have been issued, based these provisions. NBA has not, to date, itself issued any notices or fines for non-compliance. However, NBA has filed a complaint in a judicial court against a company for violation of ABS provisions in the Biodiversity Act.

The Indian Patents Act (as amended in 2005) establish disclosure of origin requirements for inventions involving biological material and the possibility of opposition and revocation of the patent for lack of compliance. The Indian Patent Office has, additionally, issued guidelines that outline the processing of such patent applications to ensure compliance with ABS requirements. NBA is in the process of filing opposition on the grant of several patents in the Indian Patent Office. NBA also has filed third party observations in foreign patent offices on applications using Indian biological resources in the inventions.

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